

SEDAR 32A: Gulf of Mexico Menhaden

Data/Assessment Workshop

Throughout the course of the DW and AW, a number of items were identified as important research topics for future stock assessments. The assessment panel evaluated the various items and developed a consensus priority list.

DATA ELEMENT	RECOMMENDATION	PRIORITY
FISHERY-INDEPENDENT ADULT INDEX	Collect Gulf menhaden ageing structures (scales and otoliths) from alternate fishing gears (e.g., gill nets and trawls) to determine gear selectivity. Need to expand efforts to age menhaden by state agencies. Determine readability of whole versus sectioned otoliths.	Very High
FISHERY-INDEPENDENT ADULT INDEX	Improve species identifications at the periphery of the Gulf menhaden's range in Texas and Alabama/Florida waters.	Very High
GENETICS AND STOCK STRUCTURE	Identification of menhaden-specific nuclear DNA markers (preferably microsatellites or SNP's) using a lab-based DNA library screening techniques. Evaluation of these markers for use in genetic studies of Gulf menhaden	Very High
FISHERY-DEPENDENT SURVEYS	A Gulf-wide aerial survey would be a useful tool to measure adult gulf menhaden abundance; "ground-truthing" for fish size and age and school size, would be a necessary adjunct to the survey	High
FISHERY-DEPENDENT SURVEYS	Additional sampling needs to be conducted to address the homogeneity of the catch in the hold of the reduction fishery vessels at the four Gulf menhaden factories. Supplemental samples must be pulled from throughout the fish hold during the pumpout process to determine if the assumption that the traditional 'last set of the trip' accurately represents the age composition for the catch for the given port-week	High
FECUNDITY/MATURITY	The seminal study on fecundity and sexual maturity of Gulf menhaden was published thirty years ago (Lewis and Roithmayr 1981) with data from the late 1970s. It is recommended that a study should be initiated to re-examine the reproductive biology of gulf menhaden in the northern Gulf of Mexico, which includes updating fecundity estimates, maturity schedules (GSI), and sex ratios. Any study needs to reinvestigate whether gulf menhaden are determinant or indeterminant spawners. Survey necessarily needs to include spawning from winter collections.	High

FISHERY-INDEPENDENT JUVENILE INDEX	Improve species identifications at the periphery of the Gulf menhaden's range in Texas and Alabama/Florida waters.	High
GENETICS AND STOCK STRUCTURE	Identification in the Clupeid literature of potential new heterologous nuclear DNA markers (preferably microsatellites or SNP's) which will potentially enhance genetic sampling in Gulf menhaden.	High
GENETICS AND STOCK STRUCTURE	Reassessment of Gulf menhaden throughout its range using a larger, more informative genetic panel of markers than that described in Anderson (2006).	High
FISHERY-INDEPENDENT JUVENILE INDEX	Design and implement a survey dedicated to determining menhaden recruitment in the coastal rivers and upper bays of the northern Gulf of Mexico.	Med/High
FISHERY-INDEPENDENT ADULT INDEX	Need to develop/expand menhaden sampling protocols for gill nets and trawls in inshore waters Standardize protocols and gears across states.	Med/High
MODELING	Benchmarks – Develop procedures to establish assessment benchmarks (e.g., Fmsy or proxies) that account for the multiple priorities of ecosystem management; such as an alteration of the calculation of Fmsy that includes predation mortality as a component of ecological yield separate from other forms of natural mortality.	Med/High
FISHERY-DEPENDENT SURVEYS	Develop fish spotter plane survey to estimate relative abundance of adult gulf menhaden; incorporate search time/flight path into survey as potential survey effort value	Medium
TAGGING STUDY	Re-visit the historical Gulf menhaden tag/recovery study. Replicate the study using 21st century tag/recapture technology. Potential products include better estimates of natural mortality, migration, growth, etc which are inputs for the stock assessment.	Medium
FISHERY-INDEPENDENT ADULT INDEX	Develop side-by-side gear comparisons among the states for standardization (trawls and gill net/strike nets).	Low/Med
PREDATOR/PREY	Expand the diet and stable isotope database to determine the trophic role of Gulf menhaden in the GOM. Investigate fatty acids profiles as an additional more specific indicator of important prey items of Gulf menhaden.	Low/Med
PREDATOR/PREY	Need to initiate food habits of major predator species in the northern GOM to determine the importance of menhaden in the diets of fish, seabirds, and marine mammals.	Low/Med
FISHERY-INDEPENDENT JUVENILE INDEX	Expand state independent sampling to include more sites in under-represented areas (Perdido Bay, Florida Panhandle, Mississippi Sound) on a monthly schedule.	Low

FISHERY-INDEPENDENT JUVENILE INDEX	Develop side-by-side gear comparisons between the states for standardization (seines and trawls).	Low
MODELING	Additional research into simulation models such as MSVPAs, ECO-SIM, EcoPath, etc.; results could produce better estimates of natural mortality as well as other fishery parameters.	Low
MODELING	Develop a habitat index to examine the potential shift in the Gulf menhaden population to more inshore waters as marsh converts to open water from coastal land loss.	Low

Review Workshop

The RW panel suggested there should be evaluation of the utility of using ovarian egg number as a proxy for SSB and notes that this will depend not only on biological considerations but also on ageing validation and errors, and selectivity determination. Ultimately, the utility of egg numbers versus SSB will depend on how status benchmarks and control rules are determined.

The Louisiana gillnet survey used in the menhaden assessment has a number of different mesh sizes and concern was expressed about developing a single index over these different mesh sizes, especially given the length frequencies presented in the assessment (AW Report, Fig. 5.44). The RW panel recommends evaluating the efficacy of developing separate indices by mesh or accounting for the different mesh sizes within the same index.

The panel did not see value in undertaking genetic studies to further elucidate Gulf menhaden population structure given the fishery operates in the center of the species distribution and it is unlikely that information gained would justify the expense of additional analyses. However, the RW panel did see considerable benefit in using simpler genetic techniques, such as DNA barcoding, to aid species identification, which is currently problematic in fishery-independent surveys conducted in peripheral range areas in Texas, Alabama, and Florida.

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GENETICS AND STOCK STRUCTURE	Identification of menhaden-specific nuclear DNA markers (preferably microsatellites or SNP's) using a lab-based DNA library screening techniques. Evaluation of these markers for use in genetic studies of Gulf menhaden	Low

GENETICS AND STOCK STRUCTURE	Identification in the Clupeid literature of potential new heterologous nuclear DNA markers (preferably microsatellites or SNP's) which will potentially enhance genetic sampling in Gulf menhaden.	Low
GENETICS AND STOCK STRUCTURE	Reassessment of Gulf menhaden throughout its range using a larger, more informative genetic panel of markers than that described in Anderson (2006).	Low

The panel provided the following comments on the research recommendations that given in the assessment documents.

Several issues were identified with ageing for menhaden including the lack of formal protocols for inter-reader comparisons and calibration/reference data sets. Given the short-lived nature of the fish, reader error of even one year can cause substantial bias in an age-based assessment. Given the pending retirement of the single ager, assessment of the accuracy of ageing and the establishment of formal protocols should be done as soon as possible.

It was not apparent to the panel that stock structure was an issue in the stock assessment and the panel did not see value in undertaking genetic studies on stock structure. However, the panel did see considerable benefit in using simpler genetic techniques such as DNA barcoding to aid species identification, which is currently problematic in peripheral range areas as sampled in the Texas, Alabama, and Florida surveys. Resolution of species identification and any other measures to ensure more consistency across the many state surveys that were excluded from the assessment could provide a more representative basis for monitoring abundance.

The recommendation to consider an aerial survey should be pursued, although the turbid waters close to the Mississippi may limit detectability of fish schools. This kind of survey offers an opportunity to form a partnership between the states, federal government and the fishing industry in a monitoring program to ensure sustainability.

The panel recommended that addressing the sampling of the catch throughout the holds of the reduction fishery vessels be rated as very high priority given concerns about the selectivity of larger fish to the catch. The 2012 study indicated that sampling only the top of the hold may underestimate the proportion of older fish in the catch and given the use of fecundity for spawning stock biomass result in an underestimate of productivity (see below).

While the studies proposed to update knowledge about the reproductive biology of Gulf menhaden would be nice to do, the panel felt that the current approach is adequate for now and more priority should be given to resolving the selectivity pattern of older fish to the fishery so that their reproductive contribution to the population can be better accounted for.